asked, “Was the world wavefunction waiting to jump for thousands of millions of years until a single-celled living creature appeared? Or did it have to wait a little longer for some more highly qualified measurer—with a Ph.D.?” If measurement has nothing to do with living observers, then what does it involve? Does it just mean that a small object, governed by quantum physics, has interacted with a big one, which is somehow exempt from quantum physics? In that case, doesn’t that mean that measurements are happening basically all of the time, and the Schrödinger equation should almost never apply? But then why does the Schrödinger equation work at all? And where’s the divide between the quantum world of the small and the Newtonian world of the large?

Finding this Pandora’s box of weird questions lying at the heart of fundamental physics is disturbing, to say the least. Yet despite all this weirdness, quantum physics is wildly successful at describing the world—much more so than simple old Newtonian physics (which was already pretty good). Without quantum physics, we wouldn’t have any understanding of why diamonds are so hard, what atoms are made of, or how to build electronics. So wave functions, with their numbers scattered across the universe, must somehow be related to the everyday stuff we see around us in the world, otherwise quantum physics wouldn’t be any good at making predictions. But this makes the measurement problem even more urgent—it means there’s something about the nature of reality that we don’t understand.

So how should we interpret this strange and wonderful theory? What story is quantum physics telling us about the world?

Rather than answering that question—which seems like it would be difficult—we could deny that it’s a legitimate question at all. We can claim that making predictions about the outcomes of measurements is all that matters in quantum physics. Now we don’t have to worry about what’s happening when we’re not making measurements, and all these difficult questions melt away. What is the wave function? How is it connected to the objects in the world around us? Easy, comforting solutions are at hand: the wave function is merely a mathematical device, a bookkeeping tool to allow us to make predictions about measurements. And it has no connection to the world around us at all—it’s merely a useful piece of mathematics. It doesn’t matter that wave functions behave differently when we’re not looking,